REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Free Fall Practice Questions

Grade 9 Advanced | Module 3 Lesson 3 | Free Fall

Name: _____ Class: _____ Date: _____

Answer all 30 questions. Select one option for each MCQ. Show equations, working, units and explanations for written questions.

Useful equations and guidance: Take upward as positive unless specified: $a = -g$; $g = 9.8\ m/s^2$; $v_f = v_i - gt$; $\Delta y = v_i t - \frac{1}{2}gt^2$.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 9 book, printed pages 71–76.

P01. Free fall means motion under:

A. No force at all B. Only air resistance
C. A constant upward force D. Gravity as the only significant force

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P02. At the highest point of an ideal vertical throw, acceleration is:

A. Downward with magnitude g B. Zero
C. Upward with magnitude g D. Dependent on the ball's speed

P03. Neglecting air resistance, objects of different masses near Earth have:

A. Acceleration proportional to mass B. The same free-fall acceleration
C. Zero acceleration D. Different acceleration solely because of mass

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P04. With upward positive, free-fall acceleration is:

A. $+9.8\ m/s^2$ always B. 0 at the top
C. $-9.8\ m/s^2$ D. $-9.8\ m/s$

P05. When an ideal dropped object falls, its speed:

A. Remains zero B. Decreases uniformly
C. Depends only on its colour D. Increases uniformly with time

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P06. An object is dropped from rest. Neglect air resistance and use downward as positive. Find its velocity after 1 s. Use $g = 9.8\ m/s^2$.

A. 19.6 m/s B. 4.9 m/s
C. 9.8 m/s D. -9.8 m/s

P07. An object falls from rest for 1 s. Neglect air resistance. Find the distance fallen using $g = 9.8\ m/s^2$.

A. 9.8 m B. 2.45 m
C. -4.9 m D. 4.9 m

P08. A ball is thrown vertically upward at 19.6 m/s. Neglect air resistance. Find the time to its highest point using g = 9.8 m/s².

A. 2 s B. 4 s
C. 1 s D. -2 s

P09. A ball is thrown upward at 19.6 m/s. Find its maximum rise. Neglect air resistance; g = 9.8 m/s².

A. 39.2 m B. 19.6 m
C. 9.8 m D. -19.6 m

P10. A stone is dropped from height 19.6 m. Calculate its impact speed. Neglect air resistance; g = 9.8 m/s².

A. 39.2 m/s B. 9.8 m/s
C. 19.6 m/s D. -19.6 m/s

P11. Define free-fall acceleration.

P12. Why is a falling feather not usually a good example of ideal free fall in air?

P13. Describe velocity and acceleration while a thrown ball rises.

P14. Describe velocity and acceleration after the highest point.

P15. For return to the launch height, compare ideal upward and downward speeds.

P16. An object is dropped from rest. Neglect air resistance and use downward as positive. Find its velocity after 1.5 s. Use g = 9.8 m/s².

P17. An object is dropped from rest. Neglect air resistance and use downward as positive. Find its velocity after 2 s. Use g = 9.8 m/s².

P18. An object falls from rest for 1.5 s. Neglect air resistance. Find the distance fallen using g = 9.8 m/s².

P19. An object falls from rest for 2 s. Neglect air resistance. Find the distance fallen using g = 9.8 m/s².

P20. A ball is thrown vertically upward at 29.4 m/s. Neglect air resistance. Find the time to its highest point using g = 9.8 m/s².

P21. A ball is thrown vertically upward at 39.2 m/s. Neglect air resistance. Find the time to its highest point using g = 9.8 m/s².

P22. A ball is thrown upward at 29.4 m/s. Find its maximum rise. Neglect air resistance; g = 9.8 m/s².

P23. A ball is thrown upward at 39.2 m/s. Find its maximum rise. Neglect air resistance; g = 9.8 m/s².

P24. A stone is dropped from height 44.1 m. Calculate its impact speed. Neglect air resistance; g = 9.8 m/s².

P25. A stone is dropped from height 78.4 m. Calculate its impact speed. Neglect air resistance; g = 9.8 m/s².

P26. A ball is thrown upward at 19.6 m/s. Neglect air resistance. Find time to the top and maximum rise.

P27. A student says gravity switches off at the highest point. Explain the error.

P28. Two balls of equal size but different masses are dropped in a vacuum. Predict their motion.

P29. Sketch a velocity–time graph for a ball thrown upward, taking upward as positive.

P30. Explain why equal times during a drop do not correspond to equal distances fallen.

